CHAPTER 4

Polynomial equations

Quadratics in $\mathbb{C}$, factorisation, number of roots

Objectives. Solve in $\mathbb{C}$ a quadratic equation with real coefficients; factorise a polynomial by $z - a$ when a is a root; solve a cubic equation when one root is known; know that a polynomial of degree n has at most n roots.

Prerequisites. Quadratics in $\mathbb{R}$ (completed square form, discriminant, sum and product of the roots); chapters 1 and 2; sum of the terms of a geometric sequence.

Fluency

Fluency · No calculator, five minutes.

1. Compute the discriminant of $2x^2 - 3x + 5$.
2. Factorise $x^2 - 5x + 6$.
3. Expand $(x - 2)\left(x^2 + 2x + 4\right)$.
4. Compute $P(2)$ for $P(x) = x^3 - 3x^2 + 4$.
5. Write $1 + q + q^2 + \cdots + q^n$ as a quotient, for $q \neq 1$.
6. Give the sum and the product of the roots of $x^2 - 7x + 10$.